

4.6 Relational Schema Development

ریلیشنل اسکیم ڈیٹا بیس کا لکھا ہوا نقشہ ہے: کون سے ٹیبل، کون سے فیلڈ، کون سی کیز، کون سے تعلقات۔

CS XI

Chapter 4: Data and Analysis

Lecture 08 of 15

4.6 Relational Schema Development

LEARN IT

A **relational schema** is the formal written design of the database: table names, fields, primary keys, foreign keys and how tables relate. You write it *before* clicking around in Access.

It defines (1) how data is split into tables, (2) which columns each table has, (3) which fields are PK/FK, (4) the relationships. Benefits: less redundancy, integrity planned in advance, easier to grow later.

WORKED EXAMPLE

Written schema for a tiny college:

```
Department (DeptID PK, DeptName)
Student     (RollNo PK, Name, Class, DeptID FK → Department)
Course      (CourseID PK, Title, Credit, DeptID FK → Department)
```

REMEMBER IT

Skipping the schema is like building a house without a drawing. You get messy, broken structures that are painful to repair.

اردو میں سمجھیں

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سافٹ ویئر کھولنے سے پہلے یہ بنانا چاہیے۔

CHECK YOURSELF

1. Name two things a schema must show.
2. Give one reason to write a schema before opening Access.

Answers: 1. Tables/fields and keys/relationships. 2. It reduces redundancy and mistakes (any valid benefit).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. A relational schema is:

- a. The blueprint of tables, keys, types
- b. A running query result
- c. A pie chart
- d. A Python list

MCQ 2. Student(roll PK, name, class) is a:

- a. Schema notation
- b. Python call
- c. OSI frame
- d. MVP

MCQ 3. Schema is designed:

- a. Before stuffing thousands of rows
- b. After losing the backup
- c. Instead of ER
- d. By the compiler

MCQ 4. Data types in schema stop:

- a. Wrong values (text in marks)
- b. TCP errors
- c. NAND delay
- d. Figma overflow

MCQ 5. Foreign keys in schema implement:

- a. Relationships
- b. Sorts
- c. Charts
- d. Sprints

MCQ 6. Changing schema later is:

- a. Painful once data exists
- b. Free always
- c. Impossible in Access
- d. A gate

Extra short questions

1. Define relational schema.
2. Write schema for Book.
3. How does schema differ from ER?
4. Name two things schema documents besides columns.

Extra long question

1. Turn your library ER into a schema listing: three tables, keys, two constraints. Justify one data type.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Formal structure: relation names, attributes, types, keys.
2. Book(ISBN PK, title, author, copies: Number).
3. ER is conceptual diagram; schema is the relational implementation.
4. PKs, FKs, unique, not null, relationships.

Long

1. Member(MemberID PK, name Text, phone Text unique). Book(ISBN PK, title Text, copies Number). Borrow(BorrowID PK, MemberID FK, ISBN FK, date Date). Copies is Number because you count them.

Next lecture: 4.7 Case Study — Library Management System